

Does ERC-8004 Form an Agent-to-Agent Trust Network?

Evidence from Early On-chain Reputation Data

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Abstract

ERC-8004 provides an on-chain identity, reputation, and validation framework for AI agents on Ethereum, yet it remains unclear whether its early ecosystem has formed a functioning agent-to-agent trust network. To address this question, this study analyzes early ERC-8004 on-chain data collected from the Ethereum mainnet from three aspects: whether ERC-8004-related on-chain activity is associated with adjusted reputation, whether reputation feedback relations are generated between agents or dominated by external accounts, and whether the registration layer is supported by active agents or diluted by zombie and batch-registered identities. The results show that ERC-8004-related on-chain activity is significantly and positively associated with adjusted reputation, suggesting that trust formation may leave measurable traces on-chain despite the off-chain nature of most agent services. However, the feedback network is still mainly driven by external accounts, with limited mutual evaluation among agents. In addition, zombie agents and batch-registered agents are widespread in the registration layer and exhibit a significant statistical association. Overall, ERC-8004 remains at an early stage and has not yet formed a mature agent-to-agent trust network. Nevertheless, it has made some ecosystem-related activities verifiable, traceable, and measurable on-chain, indicating its potential as an infrastructure for future agent trust systems.

Keywords: ERC-8004; Ethereum; AI Agent; Trust

Introduction

Blockchain provides a decentralized and tamper-resistant ledger for recording transactions, managing identity, and implementing programmable trust through smart contracts [16, 14]. With the rapid development of large language models (LLMs) and autonomous agents, agents are increasingly expected to discover services, coordinate tasks, and make decisions across open platforms. However, trustworthy collaboration among agents still requires verifiable identities, manageable permissions, traceable behaviors, and reliable reputation signals [11, 4]. These requirements make blockchain a natural infrastructure for supporting machine-readable and auditable trust relationships in emerging agent ecosystems.

The ERC-8004 standard [3] was proposed to address this need by offering a lightweight trust infrastructure for AI agents on Ethereum. It consists of an Identity Registry, a Reputation Registry, and a Validation Registry. The Identity Registry assigns each agent a unique and transferable identifier based on NFTs introduced in ERC-721 [5]; the Reputation Registry records feedback from counterparties; and the Validation Registry provides mechanisms for verifying agent outputs. The standard was proposed in August 2025 and deployed on the Ethereum mainnet in January 2026.

Existing work has mainly focused on the design and implementation of ERC-8004 [13], its integration into broader agent trust architectures [15], and the collection of on-chain agent data [7]. However, these studies have not systematically examined the early ERC-8004 ecosystem.

A key challenge is that most agent services are executed off-chain through metadata URLs, service endpoints, or external environments, while only identity, reputation, and related interactions are recorded on-

chain [7]. Therefore, Ethereum data cannot directly observe agent service quality, but it may reveal whether trust formation leaves measurable on-chain traces.

Against this background, this study asks: Has the early ERC-8004 ecosystem formed a functioning agent-to-agent trust network? To answer this question, the study examines three related aspects. First, the activity–reputation analysis tests whether reputation is connected to observable on-chain behavior, which indicates whether trust formation is traceable on-chain. Second, the feedback network analysis examines whether these reputation relations are actually generated between agents or remain dominated by external accounts. Third, the zombie and batch-registration analysis evaluates whether the registration layer is supported by active agents or diluted by inactive and speculative identities. Together, these three analyses assess whether ERC-8004 is merely an on-chain identity and reputation infrastructure, or whether it has begun to support a real agent-to-agent trust network.

Data

All data were manually collected from an Ethereum mainnet node, with collection scripts [10] and files [9] provided in the Appendix. The data cover blocks 24,339,925 to 25,277,687, spanning the period from ERC-8004 deployment on Jan 29, 2026, to the end of the observation period on Jun 09, 2026. Datasets include agent registration records, reputation feedback, on-chain transactions, and aggregated statistics.

Basic information was first recorded for all registered agents, including NFT identifier, registration time, client count, owner wallet, and bound agent wallet. The 150 agents with the highest client counts and more than two clients were selected as the core sample.

For the core sample, reputation feedback and wallet transaction records were collected. Feedback data capture the source, type, and value of each feedback transaction, while transaction records capture identity-related operations, ecosystem participation, and other on-chain activities. The data were aggregated and cleaned to compute statistics such as reputation, feedback count, transaction count, and operation-type counts, forming the basis for subsequent analyses of on-chain activity, reputation, and trust network structure.

Methodology

Activity–Reputation Relationship Measurement

Figure 1 presents the assumed causal structure, while Table 1 defines the corresponding variables.

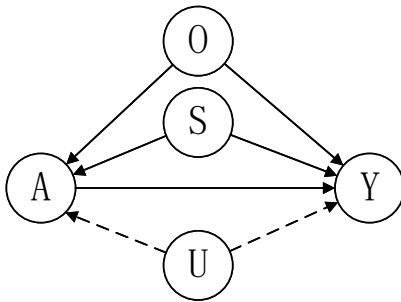


Figure 1: Causal DAG

Table 1: Notation

Symbol	Meaning	Role
A	On-chain activity	Treatment
Y	Reputation	Outcome
S	Survival time	Confounder
O	Owner Agent count	Confounder
U	Off-chain service	Unobserved

Treatment and outcome. The treatment variable, A , measures an agent’s activity within the ERC-8004 ecosystem. Agent-bound wallets may also conduct transactions unrelated to the ecosystem, such as token transfers and swaps. Counting all wallet transactions would therefore introduce substantial noise. Therefore, the activity measure is restricted to identity- and ecosystem-related operations, including agent registration, wallet setup, feedback submission, metadata updates, and agent NFT transfers. These operations are aggregated

at the agent level and log-transformed to reduce the influence of highly active agents.

$$A = \log(1 + \text{IdentityOps} + \text{EcosystemOps}).$$

The outcome variable, Y , is the adjusted reputation score. Raw reputation may be distorted by both the number of feedback records and the concentration of feedback sources. Scores based on few observations are relatively unstable, whereas scores dominated by a small number of clients may not reflect broad recognition. Raw reputation is therefore adjusted using Bayesian shrinkage [8] and then penalized for low feedback-source diversity. The resulting measure jointly reflects reputation level, feedback quantity, and source diversity.

$$Y = [\lambda \text{FeedbackScore} + (1 - \lambda) \text{FeedbackScore}] \mu, \\ \lambda = \frac{\text{FeedbackCount}}{\text{FeedbackCount} + 5}, \quad \mu = \sqrt{\frac{\log(1 + \text{ClientCount})}{\log(1 + \text{FeedbackCount})}}.$$

Confounders The analysis adjusts for on-chain survival time, S , and owner agent count, W . Survival time determines how long an agent has had to accumulate transactions and feedback. It may also capture differences in maturity and incentives to establish trust. Owner agent count captures ownership concentration and may indicate batch registration or name-squatting behavior. Both variables may simultaneously influence on-chain activity and reputation.

$$S = \log(1 + \text{SurvivalBlocks}), \quad O = \text{OwnerAgentCount}.$$

Off-chain service quality, U , is not directly observable from the available blockchain data. As illustrated in Figure 1, it may affect both ecosystem activity and reputation. Consequently, the estimated relationship should be interpreted as conditional on the observed confounders rather than as eliminating all possible sources of unobserved confounding.

Estimation strategy. Because the treatment variable, A , is continuous, the analysis uses the Generalized Propensity Score (GPS) [1] combined with Inverse Probability Weighting (IPW) [2]. The GPS represents the conditional density of observed activity given the confounders $Z = (S, O)$. It is defined as follows.

$$r = f_{A|Z}(A | Z).$$

The conditional distribution of activity is estimated using a normal treatment-assignment model. Its conditional mean is modeled as a linear function of the standardized confounders.

$$A | Z \sim \mathcal{N}(\mu(Z), \sigma^2).$$

The estimated conditional density is then used to construct stabilized inverse probability weights. The marginal density of activity forms the numerator, while the estimated GPS forms the denominator.

$$\omega = \frac{f_A(A)}{f_{A|Z}(A | Z)}.$$

To reduce the influence of extreme observations, the weights are truncated at the 1st and 99th percentiles. Weighted least squares is then used to estimate the dose-response relationship between ERC-8004 activity and adjusted reputation.

$$Y = \alpha + \beta A + \epsilon.$$

Covariate-balance diagnostics and alternative weighted specifications are used to evaluate the robustness of the results.

Feedback Network Construction

A visualization feedback network is conducted to assess whether reputation feedback in the early ERC-8004 ecosystem has formed an agent-to-agent trust network.

The network is constructed from reputation feedback transactions within the core agent sample. Each feedback record represents an evaluation submitted by one address to a specific agent and is therefore represented as a directed edge from the feedback sender to the evaluated agent [6, 12]. Nodes in the network are classified into four types: evaluated agents, smart contract addresses, agent owner addresses, and other eoa addresses. By distinguishing these node types, the visualization allows observation of whether reputation feedback primarily occurs between agents or is mostly generated by non-agent addresses.

In the visualization, agent and non-agent nodes are differentiated using distinct colors or shapes, and directed edges are drawn according to the feedback relationships.

Zombie and Batch Registration Identification

The distribution and relationship of zombie and batch-registered agents in the early ERC-8004 ecosystem also worth investigating. Zombie agents are defined as agents with no clients, or only one client without bound wallet, while batch-registered agents are those where a single owner holds more than 10 agents. To avoid misclassification of agents with insufficient observation time, the agents registered in the last 72,000 blocks (approximately 10 days) of the observation period are excluded from this analysis.

Using all agents observed during the study period, a 2×2 contingency table is constructed using the binary zombie-agent indicator and the binary batch-registration indicator.

Results

Activity-Reputation Causal Relationship

Table 2 reports the GPS/IPW weighted estimates of the activity-reputation relationship.

Table 2: GPS/IPW Weighted Estimation Results

Model	Coefficient	Standard Error	p -value	R^2	ESS
Linear model	3.0952	0.5736	< 0.001	0.1517	149.9
Quadratic sensitivity model	3.5751	0.9902	< 0.001	0.1567	149.9

The coefficient of activity score is positive and statistically significant in both models. In the linear model, a one-unit increase in activity score is associated with an average increase of about 3.10 points in adjusted reputation. The quadratic sensitivity model gives a similar marginal effect of 3.58, suggesting that the result is not driven by a strictly linear specification.

Table 3 reports the balance diagnostics for the GPS weighting.

Table 3: Balance Diagnostics for GPS Weighting

Covariate	Correlation Before Weighting	Correlation After Weighting
Log survival blocks	0.0935	0.0887
Owner agent count	-0.0231	-0.0083

The covariates are already well balanced before weighting, with absolute correlations below 0.1. After GPS weighting, the correlations move slightly closer to zero, indicating that the weighting procedure further reduces the association between the treatment variable and the observed covariates.

Overall, ERC-8004-related on-chain activity is positively associated with adjusted reputation after controlling for observable confounders. However, this result should be interpreted as a conditional association rather than definitive causal evidence. The temporal ordering between activity and reputation may not be fully clean, as some agents may become more active after receiving positive feedback. In addition, unobserved agent-level quality that cannot be directly captured from on-chain records may influence both activity and reputation.

Feedback Network

Figure 2 visualizes the ERC-8004 reputation feedback network. Each directed edge represents a feedback relation from a feedback client to an evaluated agent. The feedback network is sparse and weakly clustered. Most nodes appear as loose connections rather than dense local trust communities. The visible concentration in the upper-right part of the network is not an agent mutual-trust cluster, but is mainly caused by one central agent owner submitting feedback to many different agents.



Figure 2: ERC-8004 Reputation Feedback Network

The statistics in Figure 3 further show that the feedback network is dominated by EOA-to-agent relations, whereas agent-to-agent relations account for only a very small share of the network. Contract participation is almost absent. In addition, the comparison between raw edges and aggregated edges indicates that there are a considerable number of repeated feedback records from the same source to the same target. This repetitive feedback pattern resembles feedback-farming behavior and makes it harder for the network to develop reliable trust relations.

Overall, the feedback network provides little evidence that ERC-8004 has formed a mature agent-to-agent trust network at this early stage. Reputation feedback is observable on-chain, but the current ecosystem remains

centered on EOA accounts, which may represent human users, automated scripts, or externally controlled wallets.

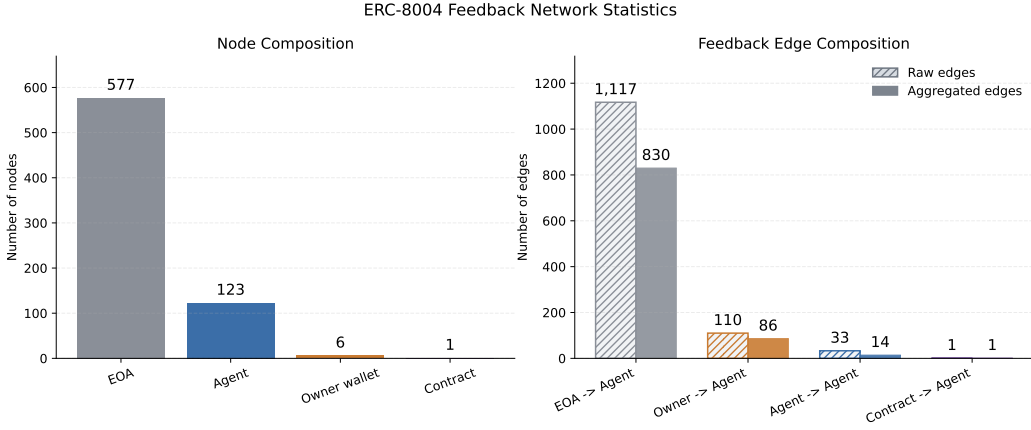


Figure 3: Composition of the ERC-8004 Feedback Network

Zombie Agents and Batch Registration Phenomenon

From the distribution of owner-held agent counts shown in the Table 4-(a), the ERC-8004 registration layer exhibits a clear structural divergence. On the one hand, a certain proportion of agents are held by owners with only one or a small number of agents. On the other hand, a large number of agents are concentrated under owners holding hundreds or even thousands of agents. This distribution suggests that early registration activity is not evenly dispersed, but instead consists of both small-scale individual registration and highly concentrated batch registration.

In terms of client count shown in the Table 4-(b), the vast majority of agents have no feedback records. Approximately 95% of agents have a client count of zero. This indicates that a large number of registered agents in the early ecosystem have not generated observable reputation interactions, and that zombie agents constitute the overwhelming majority.

Table 4: Distribution of the earliest 34,338 Registered Agents

(a) Owner-Held Agent Count			(b) Client Count		
Bucket	Agent count	Share	Bucket	Agent count	Share
1	7,367	21.45%	0	32,686	95.19%
2-10	6,111	17.80%	1	1,473	4.29%
11-50	2,716	7.91%	2	75	0.22%
51-250	4,529	13.19%	3	60	0.17%
251-1250	11,035	32.14%	4	9	0.03%
>1250	2,580	7.51%	>4	35	0.10%

Further cross-tabulation in Table 5 shows that zombie agents account for a high proportion in both the batch-registered and non-batch-registered groups. Specifically, approximately 98% of batch-registered agents are classified as zombie agents, compared with about 94% among non-batch-registered agents. The Pearson chi-square test rejects independence between zombie status and batch registration ($\chi^2 = 72.0608$, $p < 0.001$), indicating a statistically significant association between the two.

However, this result should be interpreted with caution. Since zombie agents account for an extremely large share of the overall sample, both batch-registered and non-batch-registered groups are dominated by zombie agents. Therefore, the statistical significance of the chi-square test may partly result from the large sample size and highly imbalanced category distribution, rather than necessarily implying that batch registration itself has strong explanatory power. In other words, the test result suggests that batch registration and zombie status

are not completely independent, but the more important finding is that low activity and lack of feedback have already become widespread in the early ERC-8004 registration layer.

Table 5: Zombie Agents and Batch Registration

	Batch-registered	Not batch-registered	Total
Zombie agent	20,291	12,574	32,865
Non-zombie agent	387	767	1,154
Total	20,678	13,341	34,019

Discussion

This study suggests that ERC-8004 is better understood as an on-chain identity and reputation infrastructure, rather than a mature agent-to-agent trust network.

Although a significant positive association is observed between on-chain activity and adjusted reputation, this relationship primarily reflects the alignment between observable blockchain behavior and reputation signals, rather than the emergence of endogenous trust among agents. From the perspective of the feedback network, reputation signals are still largely driven by external accounts, with very limited direct agent-to-agent evaluation. This indicates that the current ecosystem remains dominated by exogenous sources of assessment, lacking a self-sustaining trust feedback loop among agents. In addition, the prevalence of zombie agents and batch registration suggests a clear decoupling between registration and actual usage. One possible explanation is that some participants may treat agent registration similarly to NFT assets, engaging in batch registration for speculative purposes or in anticipation of future ecosystem value, rather than for immediate functional deployment. This further weakens the correspondence between registered identities and active participation.

Overall, ERC-8004 currently provides observable on-chain records of identity and reputation, but has not yet developed into an agent-driven trust network structure.

Future work may further investigate whether this registration–usage gap narrows as the ecosystem matures, and whether agent-to-agent feedback gradually becomes the dominant source of trust formation.

Conclusion

This study examines whether the early ERC-8004 ecosystem has formed a functioning agent-to-agent trust network. The results show that ERC-8004 has made agent identity, reputation feedback, and some ecosystem-related activity observable on-chain. However, the current ecosystem has not yet developed into a mature agent-to-agent trust network: reputation feedback is still mainly generated by external accounts, and the registration layer contains many inactive or batch-registered agents.

Overall, ERC-8004 should currently be understood as an early-stage infrastructure for traceable agent identity and reputation, rather than a fully functioning decentralized agent-to-agent trust network.

References

- [1] Derek W Brown, Thomas J Greene, Michael D Swartz, Anna V Wilkinson, and Stacia M DeSantis. Propensity score stratification methods for continuous treatments. *Statistics in medicine*, 40(5):1189–1203, 2021.
- [2] Stephen R Cole and Miguel A Hernán. Constructing inverse probability weights for marginal structural models. *American journal of epidemiology*, 168(6):656–664, 2008.
- [3] Marco De Rossi, Davide Crapis, James Ellis, and Erik Reppel. ERC-8004: Trustless agents. Ethereum Improvement Proposals, no. 8004, August 2025. Draft. [Online]. Available: <https://eips.ethereum.org/EIPS/eip-8004>. Accessed: Jun. 13, 2026.
- [4] Abul Ehtesham, Aditi Singh, Gaurav Kumar Gupta, and Saket Kumar. A survey of agent interoperability protocols: Model context protocol (mcp), agent communication protocol (acp), agent-to-agent protocol (a2a), and agent network protocol (anp). *arXiv preprint arXiv:2505.02279*, 2025.
- [5] William Entriken, Dieter Shirley, Jacob Evans, and Nastassia Sachs. EIP-721: Non-fungible token standard. Ethereum Improvement Proposals, January 2018. [Online]. Available: <https://eips.ethereum.org/EIPS/eip-721>. Accessed: Jun. 13, 2026.
- [6] Audun Jøsang, Roslan Ismail, and Colin Boyd. A survey of trust and reputation systems for online service provision. *Decision support systems*, 43(2):618–644, 2007.
- [7] Yulin Liu. A dataset of early blockchain-registered ai agents on ethereum. *arXiv preprint arXiv:2604.22652*, 2026.
- [8] Carl N Morris. Parametric empirical bayes inference: theory and applications. *Journal of the American statistical Association*, 78(381):47–55, 1983.
- [9] OB-David. ERC8004 Early Ecosystem data. [Online]. Available: <https://zenodo.org/records/20676023>, 2026. Accessed: Jun. 13, 2026.
- [10] OB-David. ERC8004 Early Ecosystem: Data pipeline and analysis code. [Online]. Available: <https://github.com/OB-David/ERC8004-Early-Ecosystem>, 2026. Accessed: Jun. 13, 2026.
- [11] Partha Pratim Ray. A review on agent-to-agent protocol: Concept, state-of-the-art, challenges and future directions. *Authorea Preprints*, 2025.
- [12] Paul Resnick and Richard Zeckhauser. Trust among strangers in internet transactions: Empirical analysis of ebay’s reputation system. 2002.
- [13] Ian Staley. A reference architecture for ai agents on blockchain infrastructure: Identity, policy, payments, and custody as composable primitives. *Policy, Payments, and Custody as Composable Primitives (April 25, 2026)*, 2026.
- [14] Kevin Werbach. *The blockchain and the new architecture of trust*. Mit Press, 2018.
- [15] Yunfeng Xia, Chao Li, Lei Li, Chenhao Zhang, Li Duan, Runhua Xu, and Wei Wang. Toward web 4.0: Bidirectional trust between ai agents and blockchain. *arXiv preprint arXiv:2605.08922*, 2026.
- [16] Zibin Zheng, Shaoan Xie, Hongning Dai, Xiangping Chen, and Huaimin Wang. An overview of blockchain technology: Architecture, consensus, and future trends. In *2017 IEEE international congress on big data (BigData congress)*, pages 557–564. Ieee, 2017.